# Kaden Grace – Teaching Statement

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I care about teaching so much that I sought out an adjunct position at a local community college to build more experience in the classroom. Whether at the community college or as an instructor of record at the University of Tennessee, my teaching revolves around three core concepts: personal connection, content connection, and evidence-based best practices.

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## Personal Connection

*"People don't care how much you know until they know how much you care." (Theodore Roosevelt)*

To personally connect with my students, I meet my students where they are with my authentic self. I recognize that my students are much more than just *my students*, and I do not view my students as brains on sticks. Rather, I recognize that my students carry passions, histories, and futures of their own. Knowing my students personally allows me to help shape who they become.

I present my authentic self to my students. I teach from a security in my identity that lets me admit when I make mistakes. I often jokingly remind students, "This is why I should not do algebra in front of people." By modeling humility, I can personally connect to students in an authentic way.

I ask students on the first day of the course to privately share with me a fun fact about themselves and something they passionately believe. I invite students to answer questions with a confidence that comes from knowing our classroom is a safe place to be wrong. I learn my students' names, and many of my conversations in office hours are about their hopes for their futures.

## Content Connection

*"To teach is to love a thing publicly." (Tracy McKenzie)*

I love teaching economics because it answers the *why* in life's most interesting questions. As a teacher, I connect with the content by teaching from an authentic love of the subject. I invite students to connect with the content by tying economics to the things they care about in their daily lives.

At the beginning of each day of class, I invite students to engage with the material by choosing the example good for that day. Students choose whatever is on their mind: basketballs during March Madness and protein shakes after one spilled in the classroom are some recent examples. I learn about the current events that concern my students, then I craft exam questions that ask students to use their toolkit of economic reasoning and writing to analyze these issues.

## Evidence-Based Best Practices

*"The task of the modern educator is not to cut down jungles but to irrigate deserts." (C.S. Lewis)*

Effective teaching cultivates a passion for the material, and it should be rooted in evidence-based methods. Before teaching my first class, I spent considerable time interviewing the best teachers in our department and researching effective teaching methodology. I use spaced repetition and experiential learning to help my students master economic reasoning. Spaced repetition allows the student to connect concepts throughout the course. I implement spaced repetition by beginning each lecture with a review of previous material, encouraging students to correct their written exams for half credit, and tying lectures back to earlier lectures in the course.

Passive learning, where students take notes on a lecture, often produces a superficial understanding of the material. Experiential learning, however, invites students to play an active role in the learning process through games, competitions, and simulations. In my teaching, I devote half of each class to experiential learning. I design competitions where students can gain a real-world understanding of economic concepts.

For example, I invite students to play the part of a coffee shop owner for a class period: each day, for fourteen days, the student sets a price and adjusts their recipe inputs. We debrief the game with questions like, "How did you choose your price every day? Why would you adjust your recipe?" Each student records their data, and (for extra credit) they are encouraged to estimate the structural model in the game to find the optimal recipe and price. Using what we learned from the competition, I can then teach the monopoly market structure.

Selected quotes from students associated with personal connection:

- "I have always felt like you truly cared about us learning and doing our best."
- "You are able to put yourself in our shoes, and I think this provides us with a great teaching style and understanding of our needs."
- "I like how interactive Professor Grace is. He knows all of our names but is not too pushy like some teachers can be in this aspect."
- "As a professor, you have done way better than the majority of professors I have had so far in college. I can tell you want us to succeed at your course. You encourage office hours, move at a good pace in class, and are enthusiastic, which definitely helps."
- "Greatest Haslam Professor I have ever had. Thank you for caring."
- "Amazing teacher, helps you understand and connects with the students. I would 100% take again!"

Selected quotes from students associated with content connection:

- "Whenever I ask a question in class, you are very good at explaining things in a different way than you originally did. I also ask some stupid or basic questions sometimes, and you do a good job of not making it feel like a stupid question and explaining it in a way that is easier to understand."
- "I learned a system of analytical thinking instead of a collection of concepts. Throughout this class, we got to make inferences about a business based on its data or the market circumstances. This made me reframe the way I think and approach aspects of business."
- "I learned things that can be applied to understanding current issues like tariff pass-through calculations, evaluating government policies, and other practical things."
- "My favorite part of this class was the open-ended, opinion-based reflective questions that would pop up in lectures."
- "In class examples using things we can relate to (Lululemon shirts, protein shakes, etc.) make me feel like we have an input, and it makes the learning more active."
- "I like that our class gets to pick a different product to use as an example every day; it makes the material more interesting, more applicable to the real world, and helps the information stick in my brain more."
- "I like how we are included in each class session through 'dog of the day' or just being able to choose what kind of products we want to base in-class activities on. I like how there is some humor in the class as well."
- "I enjoy how you keep lectures engaging. I have a problem with getting distracted during class, and this rarely happens in Econ because you move at the perfect pace and don't drag out lectures. At the same time, it also doesn't feel like you're trying to cram as much content as possible into one lecture. It's very balanced, and I enjoy that."

Selected quotes from students associated with evidence-based best practices:

- "My favorite aspect has been the correctable exams. There are other courses where you just finish a module and are done with the content for good, whereas here I am motivated to go back and really learn the content because I know it will show up on the final later and I will get rewarded for it."
- "I like that we get test corrections and you make it clear that what you're really focused on is us understanding the material instead of just scoring a high score on the exam. Economics is hard, but it is learnable if we get the opportunity to learn from our mistakes."
- "I enjoy the brief moment at the beginning of class where we go over what we have learned in the previous class session."

- "His lectures were easy to follow and he responded well to class feedback. He would change his lesson plans based on what the class wanted. For example, if they were struggling with a certain problem or concept, he would take extra time to go over it and give more practice."
- "My favorite part of this class was the coffee shop game."
- "I loved getting to learn through engaging games and conversation. You're an amazing professor, and I believe you have a bright future ahead!"
- "My biggest annoyance with ECON classes in general is the full class period lecture that can become boring and difficult to focus on at times. By incorporating activities and follow-along exercises, it makes it a lot more engaging."
- "Treats lessons like a puzzle to solve and not a page to memorize."
- "Well-organized lectures with engaging games/methods of teaching. Thought it was a nice mix of fun and work that made it easy to learn."

**University of Tennessee**
**Haslam College of Business**
**ECON 311 Intermediate Microeconomics**
**Spring 2026**
**On campus, in person**
**Haslam Business Building (HBB) 136**
**Tues/Thurs 4:05 pm – 5:20 pm**


**Instructor:** Kaden Grace
**Email:** kaden.grace@utk.edu OR kgrace7@vols.utk.edu
**Website:** www.kadengrace.co
**Office:** Stokely Management Center (SMC) Office 212

**Office Hours:** Tuesdays 2:00 pm – 4:00 pm in person, or email me for an appointment
**Final Exam:** Monday, May 11, 3:30 pm – 6:00 pm

# Required Materials:

1) **MyLab Economics Inclusive Access for HW and e-text:** Graded homework will be done online. Inclusive access grants you access to Pearson MyLab. In Pearson, you can find an online version of the text Microeconomics, 9th edition (2020), by Pindyck and Rubinfeld. Older editions (8th, 7th, 6th, or 5th) of the paper text may be used as a reference (basic micro theory has not changed much!). Unless you 'Opt Out', the cost has already been included in your course fees.
2) **Notebook/binder with paper and pencil/pen and/or Laptop/iPad/Surface**: These are for taking notes in class. I highly recommend pencil and paper for the examples we go over in class.
3) **Calculator:** A graphing calculator or some type of calculator other than your phone will be helpful for homework, in-class work, and exams. Please bring your calculator to class each day.
4) **Smartphone:** We may have occasional questions in class via Point Solutions (Clickers) for attendance and possible review sessions with Kahoot.

**Supplementary materials will be provided on an as-needed basis.**

**Prerequisites:** ECON 201 or ECON 207 or ECON 211 or ECON 217.
**Corequisites:** STAT 201 or STAT 207

# Course Description

This course builds upon the microeconomics framework in your introductory economics course. Many of the topics require the proper application of various mathematical tools. You will learn to apply the appropriate tools and theories to make informed predictions about market changes.

# Course Objectives

By the end of the course, students will be able to:

- Model economic phenomenon with simple supply and demand graphs.
- Demonstrate proficiency in solving simple optimization problems.

- Describe preferences and utility in a mathematical way.
- Apply mathematical optimization techniques to both consumer and producer problems.
- Describe the economic intuition of the optimality conditions for both consumers and producers.
- Compare actual consumer and producer behavior to economic theory.
- Analyze the welfare impacts of government interventions on an open market.
- Contrast how different market structures affect firm behavior and supply.
- Demonstrate strategic firm behavior in an oligopoly market structure using game theory.
- Identify classical economic theory in real-world public policy, news, or personal experience.

## Course Activities

## Graded Homework in MyLab Economics

The ten chapter homeworks in MyLab are weighted the same regardless of the number of questions, and I will drop your lowest score. MyLab questions require applying the math skills and economic theory from class. Homework will be available at least five days prior to the 11:59 pm deadline on the due date. Homeworks may be done at your leisure once they are posted, and you are encouraged to work with your classmates. Please do not, DO NOT wait until the last minute to attempt the graded homework – do it in advance to allow time to figure out the answers and address potential technical difficulties. Allow for at least 4 hours. I will also provide you with the number of tries per question for each homework, as it varies by question type (multiple-choice vs. short-answer).

## Attendance

You are required to attend class. This class is math-intensive, and it takes practice, practice, practice! To get the most out of each class, some preparation is required outside of class, and daily expectations will be posted on Canvas. Lack of preparation will reduce your attendance score. Some days, you will be asked to complete small assignments individually or in groups, and these assignments may be evaluated. Working with your classmates is especially helpful because you can learn from each other and teach one another. The best way to learn a concept is to teach it to someone else. If you miss a class with graded attendance or work, you cannot receive a make-up opportunity unless you have an official university excused absence.

## Midterm Exams

Midterm Exams are worth 100 points each. You are expected to take the exams on the scheduled dates. No makeup exams will be given unless you have an excused absence that we discuss in advance. Missed exams will be given a score of zero. There are two exams taken during the semester.

## Final Exam (cumulative)

There will be an in-person and CUMULATIVE final exam on Monday, May 11, 3:30 pm – 6:00 pm in our regularly scheduled classroom.

## IMPORTANT!
## Late Work Policy and Personal Responsibility

You and only you are responsible for completing the learning activities on time. If you miss a deadline for any course activity without my prior written permission (before the due date), there are no extensions or make-ups. If you are traveling for an officially sanctioned University activity on a due date and require accommodations, it is your responsibility to show me the University paperwork in advance. If you have a medical or family emergency, you must obtain a letter from the Dean of Students. The Dean of Students will only grant letters if you have a serious illness or extenuating family circumstances.

## Instructor Responsibility

I am responsible for facilitating the learning process for students who are committed to learning. I am responsible for determining the extent to which students have met the stated goals in this course. I am expected to be fair, courteous, and respectful, to be responsive to student needs, and not to discriminate based on race, gender, ethnicity, or sexual or religious preference. Grades are assigned based on my professional judgment of the quality of your work in accordance with the syllabus.

## Student Responsibility

I assume each student has read and signed the Honor statement regarding academic integrity in Hilltopics. Anyone in violation of the Honor statement, such as receiving inappropriate academic assistance or copying another student's work, will be subject to the highest penalty, either failing the class or expulsion. You are responsible for your learning outcomes and your performance on all assignments. You are responsible for contacting the professor early in the semester if you are struggling with the material.

## Email Etiquette

When composing an email, it is polite to include a salutation, such as "Dear Mr. Grace," as well as your own name at the end. Also, since I have many students, please include your course name (Econ 311) in the email subject. I will do my best to respond to emails within 24 hours, excluding weekends and holidays. I will only check and respond to email within business hours (8:00 am – 5:00 pm). If you do not hear a response from me within 48 hours, please resend your message.

## Grading Scale

Your course grade will be determined solely by performance. This is a 3-credit class, which means you should expect to devote ~6 hours each week (~one hour a day) to studying economics, not including class time. There will be NO "curve". Please email or talk to me EARLY in the semester if you have serious concerns about your grade for this class (i.e., for scholarships). Economics majors need a C (not a C-) for this class to count towards their major. I will do everything I can to help you better understand the material. There are also many resources listed below to help students navigate the semester.

| Assignment Point Breakdown | Percent of Grade | *Grading Criteria | |
|---|---|---|---|
| MyLab Economics HW (drop lowest) | 35% | 93−100% | A |
| Attendance | 10% | 90−92% | A− |
| Midterm Exam 1 (Ch 1,2,3,4) | 15% | 87−89% | B+ |
| Midterm Exam 2 (Ch 6,7,8,9) | 15% | 83−86% | B |
| Final Exam (cumulative) | 25% | 80−82% | B− |
| **Total** | **100%** | 77−79% | C+ |
| | | 73−76% | C |
| | | 70−72% | C− |
| | | 67−69% | D+ |
| | | 63−66% | D |
| | | 60−62% | D− |
| | | 59% or less | F |

***Using the laws of mathematics found in an Excel worksheet, I will round your final grade to the nearest point. For example, if you have a 92.5, this rounds to a 93%, or an A. If you have a 92.4% (or the dreaded 92.4999999999999%), this rounds to 92%, or an A–.**

## Course Outline (Subject to revisions. Additional details available on Canvas.)

| Week | Class Dates | Textbook Chapter | Homework/Exams |
|---|---|---|---|
| 1 | Jan 20, 22 | Math review<br>Chapter 1 Preliminaries<br>Chapter 2 The Basics of Supply and Demand | Practice Homework (extra credit) |
| 2 | Jan 27, 29 | Chapter 3 Consumer Behavior | Homework 1 (Ch 1,2) |
| 3 | Feb 3, 5 | Chapter 3 (continued) | Homework 2 (Ch 3) |
| 4 | Feb 10, 12 | Chapter 4 Individual and Market Demand | Homework 3 (Ch 4) |
| 5 | Feb 17, 19 | Chapter 4 (continued) | **Exam #1 (Ch 1,2,3,4) Tues Feb 17** |
| 6 | Feb 24, 26 | Chapter 6 Production | |
| 7 | Mar 3, 5 | Chapter 7 Production Costs | Homework 4 (Ch 6) |
| 8 | Spring Break! | Spring Break! | |
| 9 | Mar 17, 19 | Chapter 8 Profit Maximization and Competitive Supply | Homework 5 (Ch 7) |
| 10 | Mar 24, 26 | Chapter 9 The Analysis of Competitive Markets | Homework 6 (Ch 8) |
| 11 | Mar 31 (Apr 2 no class) | Chapter 9 (continued) | Homework 7 (Ch 9) |
| 12 | Apr 7, 9 | Chapter 10 Market Power: Monopoly and Monopsony | **Midterm Exam #2 (Ch 6,7,8,9) Tues April 7** |
| 13 | Apr 14, 16 | Chapter 13 Game Theory and Competitive Strategy | Homework 8 (Ch 10) |
| 14 | Apr 21, 23 | Chapter 12 Monopolistic Competition and Oligopoly | Homework 9 (Ch 13) |
| 15 | Apr 28, 30 | Chapter 12 (continued) | Homework 10 (Ch 12) |
| 16 | May 5<br>(May 7, no class - Study Day) | Chapter 15 Investment, Time, and Capital Markets, cumulative review | |
| Final | May 11 | | Final Exam: Monday, May 11, 3:30 pm (cumulative) |

## Key Campus Resources for Students:

- Center for Career Development and Academic Exploration (Career counseling and resources; Handshake job search system)
- Course Catalogs (Listing of academic programs, courses, and policies)
- Hilltopics (Campus and academic policies, procedures, and standards of conduct)
- OIT HelpDesk **(865)** 974-9900
- Schedule of Classes/Timetable
- Student Health Center (visit the site for a list of services)
- Academic Success Center (Academic support resources)
- Undergraduate Academic Advising (Advising resources, course requirements, and major guides)
- University Libraries (Access to library resources, databases, course reserves, and services)

## Copyright Statement

The instructor of this class owns the copyright to the syllabus, handouts, assignments, quizzes, and exams associated with the class. All presentations developed by the instructor, as well as the instructor's lectures, are also protected by copyright, whether these presentations and lectures are delivered live in-class, shared through Zoom or other videoconference platforms, or uploaded to Canvas or similar sites.

**Sharing any of this material without the written permission of the instructor is a violation of copyright law, and is therefore also a violation of the University's policy on acceptable use of information technology resources (UT policy number IT0110). That policy states that students will not commit copyright infringement, "including file-sharing of video, audio, or data without permission from the copyright owner," and that file sharing is a violation of the University's student code of conduct. I will report all such violations to the Office of Student Conduct and Community Standards.**

## Campus Syllabus

Dear Student,

The purpose of this Campus Syllabus is to provide you with important information that applies to all UTK courses. Please observe the following policies and familiarize yourself with the university resources listed below. At UT, we are committed to providing you with a high-quality learning experience. I want to wish you the best for a successful and productive semester.

–Dr. John Zomchick, Provost and Senior Vice Chancellor

ACADEMIC INTEGRITY

Each student is responsible for their personal integrity in academic life and for adhering to UT's Honor Statement. The Honor Statement reads: "An essential feature of the University of Tennessee, Knoxville is a commitment to maintaining an atmosphere of intellectual integrity and academic honesty. As a student of the university, I pledge that I will neither knowingly give nor receive any inappropriate assistance in academic work, thus affirming my own personal commitment to honor and integrity."

YOUR ROLE IN IMPROVING THE COURSE THROUGH ASSESSMENT

At UT, it is our collective responsibility to improve the state of teaching and learning. During the semester you may be requested to assess aspects of this course, either during class or at the completion of the class, and through the TNVoice course evaluation. Please take the few moments needed to respond to these requests as they are used by instructors, department heads, deans and others to improve the quality of your UT learning experience.

STUDENTS WITH DISABILITIES – http://sds.utk.edu

The University of Tennessee, Knoxville, is committed to providing an inclusive learning environment for all students. If you anticipate or experience a barrier in this course due to a chronic health condition, a learning, hearing, neurological, mental health, vision, physical, or other kind of disability, or a temporary injury, you are encouraged to contact Student Disability Services (SDS) at 865-974-6087 or sds@utk.edu. An SDS Coordinator will meet with you to develop a plan to ensure you have equitable access to this course. If you are already registered with SDS, please contact your instructor to discuss implementing accommodations included in your course access letter. You may also contact your SDS Coordinator if you have any questions, concerns, or experience any barriers to using your accommodations in this course.

ACCESSIBILITY POLICY AND TRAINING – http://accessibility.utk.edu

The University of Tennessee, Knoxville, provides reasonable accommodations for individual students with disabilities through its office of Student Disability Services. The university is also committed to making information and materials accessible, when possible. Resources and assistance to support these efforts can be found at http://accessibility.utk.edu/.

WELLNESS – http://wellness.utk.edu/ and http://counselingcenter.utk.edu/

The Center for Health Education and Wellness empowers all Volunteers to thrive by cultivating personal and community well-being. The Center can answer questions about general wellness, substance use, sexual health, healthy relationships, and sexual assault prevention. The Student Counseling Center is the university's primary facility for personal counseling, psychotherapy, and psychological outreach and consultation services.

The Student Health Center has a wide range of clinics and services. Their goal is to provide quality healthcare and assist you in maintaining a healthy lifestyle to support your success as a student. To schedule an appointment, call 865-974-5080. To report a health-related absence visit https://studentlife.utk.edu/caring-for-all-vols/absence/.

Any student who has difficulty affording hygiene products, groceries, or accessing sufficient food to eat every day is urged to contact Basic Needs for support. The Big Orange Pantry, located on the 3rd floor of the Student Union, is a free resource for all students, faculty, and staff, no matter how great or small their need is. Students who need emergency financial assistance can also request funding from the Student Emergency Fund.

Students who are experiencing non-academic difficulty or distress and need assistance should call 974-HELP or submit an online referral. The 974-HELP team specializes in aligning resources and support to students experiencing mental health distress.

EMERGENCY ALERT SYSTEM – http://safety.utk.edu/
The University of Tennessee, Knoxville, is committed to providing a safe environment for learning and working. When you are alerted to an emergency, please take appropriate action. Learn more about what to do in an emergency and sign up for UT Alerts. Check the emergency posters near exits and elevators for building specific information. In the event of an emergency, course schedules and assignments may be subject to change. If changes to graded activities are required, reasonable adjustments will be made, and you will be responsible for meeting revised deadlines.

*The instructor reserves the right to revise, alter or amend this syllabus as necessary. Students will be notified in writing via Canvas of any such changes.*

# Extra Credit: Solving the Coffee Shop Game

Econ 311: Intermediate Microeconomics, Spring 2026

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## Background

In our last class session, you played the Coffee Shop browser game and logged your data: recipe settings, temperature, cups sold, starting and ending cash, and reputation. The class pooled this data into a single spreadsheet. This assignment asks you to use that data and the tools from this course to *solve* the game. You will estimate how demand works, figure out what the optimal recipe is, and find the profit-maximizing price a monopolist should charge.

Every step of this assignment connects to something we covered in class: demand estimation, cost minimization, monopoly pricing, and elasticity. The data is real, which means it is messy. Part of this assignment is figuring out how to deal with that.

The structural assumptions we are making about the game are in `modeling.md`. Read that document before starting. It defines the demand equation, the reputation dynamics, how price is recovered from the data, and how optimal price is derived. You will estimate the unknown parameters in those equations using the class data.

The dataset is the CSV file from the Google Form.

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## Questions

Answer all three questions. Show your work: include your code, your regression output, and your reasoning.

**1. What is the optimal recipe?**

Estimate the recipe goodness function from `modeling.md` using the class data. Your dependent variable is per-cup reputation change ($g_t = \Delta\text{Rep}/Q$). Based on your estimates, which combination of coffee, milk, and sugar maximizes reputation goodness? Take input prices into account. The game sells ingredients in tiers (see `modeling.md`), so the optimal recipe under a tight budget may differ from the unconstrained optimum.

**2. Given the optimal recipe, my starting reputation, and tomorrow's temperature, what is the profit-maximizing price?**

Estimate the demand equation from `modeling.md`. Use your estimated coefficients and the monopoly pricing formula to compute $P^*$ for at least two scenarios: a cold day with low reputation, and a hot day with decent reputation. For each scenario, compute the expected quantity $Q^*$ and recommend which supply tier to buy.

**3. What is the price elasticity of demand? How does this affect the optimal markup?**

Compute the elasticity using your estimated demand slope and the sample mean price and quantity. Interpret the number: is demand elastic or inelastic at the mean? Compute the Lerner Index. Does the model's implied markup make intuitive sense given how the game works?

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## Notes

**On using AI tools:**

You are allowed -- and encouraged -- to use generative AI (Claude Code, ChatGPT, Codex, etc.) to help you write code and plan your analysis. This is not just permitted; it is realistic practice for how empirical work gets done. If you use AI, note briefly which tool you used and what it helped you with.

That said, you are responsible for understanding what your code does and why. If your regression output says something strange, you need to be able to explain what went wrong.

**On simultaneous equations bias:**

There is a problem lurking in the demand estimation called simultaneity bias. It comes up because students who expected high demand tended to charge higher prices. The price and the error term in your demand equation are not independent. OLS assumes they are, so the estimated price coefficient will be biased.

You do not need to fully fix this problem, but you do need to address it. Some things to think about:

- What sign is the bias? Does OLS overestimate or underestimate the effect of price on demand?
- Is there a way to reduce (if not eliminate) this bias using the structure of the panel? One approach: instead of comparing students to each other, compare a given student's day 2 to day 1. Why might this help?
- Do you have a valid instrument for price? A valid instrument has to affect price but have no direct effect on demand. Is temperature a valid instrument? Is marginal cost?

At a minimum, describe how you think the bias is affecting your estimated elasticity and optimal price.

**On the data:**

The data was collected by your classmates and is not perfectly clean. Some things to watch for:

- The price formula (ending cash minus starting cash, divided by cups sold) will give nonsense answers for some rows -- negative prices, prices above $10, etc. You will need to decide what to do with these observations.
- Reputation was logged in multiple formats: some students entered numbers, some entered text, some left it blank. You will need to decide how to handle each case.
- Some students entered starting and ending cash in the wrong order. If the formula gives a negative price but swapping the two columns gives a reasonable price, that is probably what happened.

In each case, document your cleaning decisions.

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## What to Submit

A single document (PDF or Word) containing:

1. A brief description of how you cleaned the data and why
2. Your regression output for the goodness model and the demand model (a table or screenshot is fine)
3. Answers to each of the three questions above, with your reasoning
4. A note on simultaneity bias and how it affects your results

5. Your code (in an appendix or a separate file)

You do not need to match my results exactly. I only care that you give a thoughtful analysis with well-reasoned choices.